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HR Analytics: Employee Attrition and Retention Insights Dashboard using Power BI

Project Domain

[Excel / Power BI]

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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Year - 2026

3. Problem Statement

Employee attrition is a major challenge faced by organizations, as it directly impacts productivity, employee morale, and overall business performance. Organizations generate large volumes of HR data, including employee demographics, job roles, salaries, performance metrics, and work experience. However, this data is often raw and underutilized due to lack of structured analysis and meaningful insights.

This project focuses on the domain of Human Resource Analytics, using the IBM HR Analytics Employee Attrition & Performance dataset obtained from Kaggle**.** The dataset contains detailed employee information such as age, gender, department, job role, monthly income, job satisfaction, education, work experience, and attrition status. Although structured, the dataset requires data cleaning, preprocessing, and analytical evaluation to identify patterns and key factors influencing employee attrition.

The problem is significant in real-world organizations because high attrition leads to increased recruitment costs, loss of skilled employees, and operational inefficiencies. Without proper analysis, organizations are unable to identify the root causes of employee turnover, such as low job satisfaction, inadequate compensation, or lack of career growth.

This project addresses the gap between raw HR data and actionable insights by applying data cleaning, KPI generation, and visualization techniques using Power BI. The solution supports HR managers and decision-makers in identifying attrition trends and developing effective employee retention strategies through data-driven decision-making.

4. Objectives

• To collect and explore the IBM HR Analytics dataset from Kaggle and understand its structure, key attributes, and business relevance.

• To clean and preprocess the dataset by removing duplicates, handling missing values, and standardizing categorical and numerical fields.

• To create derived metrics such as Attrition Rate, Average Tenure, Salary Distribution, and Department-wise Attrition.

• To analyze employee attrition based on factors such as job satisfaction, salary, work experience, and department.

• To calculate key performance indicators (KPIs) including Total Employees, Attrition Count, Attrition Rate, and Average Monthly Income.

• To identify patterns and trends in employee turnover across different departments, job roles, and experience levels.

• To develop an interactive dashboard using Power BI that visually represents attrition insights through charts, KPIs, and filters.

• To transform raw HR data into meaningful insights that support better employee retention strategies and HR decision-making.

• To document the complete workflow including data cleaning, analysis, dashboard development, and project files for submission.

5. Methodology / Implementation Plan

#	Phase	Description
1	**Data Collection & Exploration**	The IBM HR Analytics Employee Attrition & Performance dataset was obtained from Kaggle. It contains employee information such as age, department, job role, income, job satisfaction, work experience, and attrition status. Initial exploration involves reviewing dataset structure, understanding columns, and validating data types.
2	**Data Cleaning & Preprocessing**	The dataset was cleaned using Excel and Power BI tools. Duplicate records were removed, missing values were handled, and categorical and numerical fields were standardized. Data consistency was ensured to enable accurate analysis.
3	**Analysis & AI Integration**	Data analysis was performed using Power BI DAX measures to calculate key metrics such as Attrition Rate, Total Employees, Average Income, and Department-wise Attrition. AI tools such as Microsoft Copilot or ChatGPT were used to generate summaries and interpret insights.
4	**Visualization / Output**	An interactive Power BI dashboard was developed using bar charts, pie charts, KPI cards, and trend visuals. The dashboard displays Attrition Rate, Employee Distribution, Salary Analysis, and Department-wise Attrition with slicers for dynamic filtering.
5	**Documentation & Submission**	The complete workflow including dataset source, data cleaning, analysis, dashboard development, and AI-generated insights was documented. All files were organized and prepared for submission along with GitHub repository and report.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop — Intel(R) Core(TM) i3-7020U CPU (8 GB RAM)
- Operating System: Windows 11
- Microsoft Excel (Microsoft 365)

- Power BI Desktop — latest version
- Microsoft Copilot — via Microsoft 365 subscription
- AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — for prompt-based projects
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge

B. Dataset

• The dataset used in this project is the IBM HR Analytics Employee Attrition & Performance dataset, obtained from Kaggle.

• It contains employee information such as age, gender, department, job role, monthly income, job satisfaction, education, work experience, and attrition status.

• After downloading the dataset, data cleaning and preprocessing were performed to ensure accuracy and consistency.

• Duplicate records were checked, missing values were handled, and categorical data was standardized.

• Additional calculated fields such as Attrition Rate, Experience Groups, and Income Categories were created to enhance analysis.

• These transformations helped in identifying key factors influencing employee attrition and enabled effective dashboard visualization in Power BI.

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

• A clean and validated HR dataset with all preprocessing steps documented.

• Creation of key HR metrics such as Attrition Rate, Average Salary, and Department-wise Attrition.

• An interactive Power BI dashboard displaying employee distribution, attrition trends, and key influencing factors.

• Identification of major causes of employee turnover to support better retention strategies.

• Use of AI tools such as Microsoft Copilot or ChatGPT to generate analytical summaries and insights.

• A structured HR analytics solution that transforms raw employee data into actionable insights for improved decision-making.

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